

4	(a)	product of the force and the perpendicular distance to/from a point/pivot	B1	[1]
	(b)	(i)		
		$4000 \times 2.8 \times \sin 30^\circ$ or $500 \times 1.4 \times \sin 30^\circ$ or $T \times 2.8$ or 4000×1.4 or 500×0.7	B1	
		$4000 \times 2.8 \times \sin 30^\circ + 500 \times 1.4 \times \sin 30^\circ = T \times 2.8$ hence $T = 2100$ (2125) N	M1 A0	[2]
		(ii)		
		$(T_v = 2100 \cos 60^\circ =) 1100$ (1050) N	A1	[1]
		(iii)		
		there is an upward (vertical component of) force at A	B1	
		upward force at A + T_v = sum of downward forces/weight+load/4500 N	B1	[2]

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